SRI LANKA INSTITUTE OF INFORMATION TECHNOLOGY

Faculty of Computing — Department of Software Engineering

Research Project — Periodic Log Entry Form

Project **R26-SE-009** | Agent 3 | **Hamna Hakeem** (IT22516916) | Reporting period **23.03.2026 – 25.08.2026**

1. PROJECT AND STUDENT DETAILS

Project ID	R26-SE-009
Research Topic	Self-Optimized LLM Multi-Agent System for Software Engineering
Supervisor	Prof. Nuwan Kodagoda
Co-Supervisor	Mr. Eishan Weerasinghe
Student Name	Hamna Hakeem
Student Registration No.	IT22516916
Individual Component	Agent 3 — QA Tester (AI QA Engineer) and the QA user interface
Reporting Period	23 March 2026 to 25 August 2026
Phase Covered	Requirement, implementation and integration phase, representing approximately 65% of the planned project work.

2. SUMMARY OF RP WORK CARRIED OUT DURING THE REPORTING PERIOD

This reporting period covers the development of Agent 3, the quality assurance and validation component of the multi-agent system, together with the QA-facing user interface. The period began by structuring the QA workflow — unit testing, integration testing, API contract validation, quality scoring, issue classification and round-based feedback to Agent 2 for corrective action — and by implementing the first test routines, the report writer and the run endpoints. The QA interface followed: the cycle timeline, bug reporting and contract review screens, evidence and artefact views, memory logs and project history, together with the analyser and dependency graph used when preparing a generated project for testing. Evaluating that first version showed that staged demonstration testing had to be replaced by real execution, which shaped a redesign into distinct unit, runtime and API, end-to-end and security stages, each bound to stored evidence. The later part of the period implemented that design: test authoring with integrity guards, a repair loop for failing tests, an installable harness with mock handling, and isolated per-run sessions and snapshots.

3. DATE-WISE RP WORK (43 ENTRIES)

MARCH 2026

4 ENTRIES

NO.	DATE	RP WORK CARRIED OUT
1	23.03.2026	Defined the scope of Agent 3 as the AI QA tester that validates generated systems and returns actionable defects to the builder.
2	26.03.2026	Designed the QA workflow covering unit testing, integration testing, API contract checks, scoring and bug reporting.

NO.	DATE	RP WORK CARRIED OUT
3	28.03.2026	Identified the test stages and reporting outputs needed for a repeatable quality gate before deployment.
4	31.03.2026	Defined issue classification with severity levels, module mapping and fix recommendations.

APRIL 2026

8 ENTRIES

NO.	DATE	RP WORK CARRIED OUT
5	03.04.2026	Defined the quality score structure covering functional quality, security, performance and requirement coverage.
6	07.04.2026	Designed the feedback loop to Agent 2 so a failed run triggers guided repair instead of manual debugging.
7	10.04.2026	Designed round-based re-testing so verified fixes do not require the whole cycle to be repeated.
8	14.04.2026	Planned the report artefacts — QA summaries and issue bundles — consumed by the rest of the team.
9	17.04.2026	Implemented the unit-testing and integration-testing routines for generated projects.
10	21.04.2026	Implemented the PDF report writer used to publish QA output.
11	24.04.2026	Implemented the test-run start and status endpoints with progressive logs and summary metrics.
12	28.04.2026	Agreed with the group that QA pass status acts as the deployment gate for Agent 4 and surfaces on the shared dashboard.

MAY 2026

8 ENTRIES

NO.	DATE	RP WORK CARRIED OUT
13	04.05.2026	Set up the QA front-end routes and the base screen structure of the testing workspace.
14	07.05.2026	Built the QA cycle timeline and the testing progress states.
15	11.05.2026	Built the bug-reporting and contract-review screens used to present QA findings.
16	14.05.2026	Revised the bug-reporting and contract-review screens after the first internal review round.
17	18.05.2026	Built the artefacts and evidence presentation screens for test output.
18	21.05.2026	Added the memory-log and project-history views so previous QA runs stay visible to the user.
19	25.05.2026	Added the shared QA components and wired the QA routes into the common navigation.
20	28.05.2026	Added the Node.js analyser and dependency graph used when preparing a generated project for testing.

JUNE 2026

8 ENTRIES

NO.	DATE	RP WORK CARRIED OUT
21	02.06.2026	Reviewed the first QA cycle and identified where staged demonstration testing had to become real execution.

NO.	DATE	RP WORK CARRIED OUT
22	05.06.2026	Ran the existing suite against generated projects and recorded why the results could not yet be trusted.
23	09.06.2026	Compared test-harness and runner options for generated JavaScript and React projects.
24	12.06.2026	Designed the redesigned QA stages covering unit, runtime and API, end-to-end and security verification.
25	16.06.2026	Defined the evidence model so every verdict is backed by a stored artefact rather than a log line.
26	19.06.2026	Specified the unit-test authoring loop that derives tests from the specification and the generated code.
27	23.06.2026	Specified the repair loop so failing generated tests are corrected instead of discarded.
28	26.06.2026	Defined the author guards that prevent a test being weakened simply to make it pass.

JULY 2026

8 ENTRIES

NO.	DATE	RP WORK CARRIED OUT
29	01.07.2026	Designed the end-to-end stage: preflight checks, console capture, normalisation and grounding of observed behaviour.
30	06.07.2026	Specified the API verification stage and the contract checks applied to generated endpoints.
31	09.07.2026	Specified the security checks and the point in the staged run at which they are applied.
32	13.07.2026	Designed per-run session isolation so concurrent test runs cannot interfere with one another.
33	16.07.2026	Designed the snapshot mechanism that records project state before and after each repair round.
34	21.07.2026	Reviewed the integration points with the builder so failures return actionable repair context.
35	24.07.2026	Defined the QA handoff artefacts Agent 4 requires so deployment gating can be enforced.
36	29.07.2026	Reviewed the rebuilt QA design with the group and closed the remaining open points.

AUGUST 2026

7 ENTRIES

NO.	DATE	RP WORK CARRIED OUT
37	04.08.2026	Implemented the unit stage: specification derivation, test authoring and the execution runner.
38	07.08.2026	Implemented the author repair path together with the guards that protect test integrity.
39	12.08.2026	Implemented the runtime and API verification stage over the shared runs service.
40	15.08.2026	Added the QA agent component and the testing interface into the shared Studio.
41	19.08.2026	Implemented the end-to-end preflight, console capture and normalisation stages.
42	22.08.2026	Split the test-harness foundations into reusable modules and added harness installation support.
43	25.08.2026	Implemented mock handling and split the session foundations so each run keeps isolated state and snapshots.

4. PROJECT WORK PLANNED BEFORE THE NEXT EVALUATION PERIOD

1. Complete the end-to-end stage — preflight, console capture, normalisation and grounding — and converge it against real generated applications.
2. Finish the unit-test authoring and repair loop so failing generated tests are corrected rather than discarded.
3. Complete API verification and the security checks, and bind every verdict to stored evidence.
4. Improve the repair-feedback loop so Agent 2 receives clearer defect context and regression indicators.
5. Tighten the QA handoff artefacts consumed by Agent 4 so deployment gating is stricter and more reliable.

5. DECLARATION AND SIGNATURES

I certify that the work recorded in this log book for the period 23 March 2026 to 25 August 2026 was carried out by me as the individual component contribution to research project R26-SE-009, and that the group-level contributions recorded here were made jointly with the other project members.

SUPERVISOR'S COMMENTS

Hamna Hakeem STUDENT — IT22516916 Date: ______________	**Prof. Nuwan Kodagoda** SUPERVISOR Date: ______________	**Mr. Eishan Weerasinghe** CO-SUPERVISOR Date: ______________